Technology fields, automation and task reinstatement

Working paper

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Abstract

We use the geometric task-space framework of Storck and Andersson (2026) to study how the reach of a technology determines whether task creation benefits or harms workers in different parts of the labour market. The framework places occupations as task bundles in a two-dimensional task space calibrated from occupational embeddings. The market price of skill is a field over this space, directional rather than uniform, so that depth is rewarded in the socio-cognitive directions and not in the service and manual directions. Automation removes the high-priced tasks from a bundle; reinstatement returns a fraction γ of the displaced mass as new tasks, which attach to an occupation only where the priced capability they require is already held. Occupations are therefore redefined rather than abolished, and their wages move with the priced content of their bundles. New tasks that no occupation can attach remain unbound, and their concentration marks where new occupations may emerge.

Capital bears a task only where it is the cheaper producer, so automation takes the dearest work first; the per-unit cost saving it yields vanishes at the adoption margin and grows only deep in a technology's core. We close the model with competitive pricing: the saving is passed to consumers as a lower price, and demand of elasticity η scales each place's revenue, drawing labour into a cheapened direction when demand is elastic and releasing it when demand is inelastic. Because the saving is second-order at the margin, displacement dominates at plausible elasticities; net employment growth in an automated direction requires either elastic demand or large infra-marginal saving. The mechanism is the spatial form of the demand channel of Bessen (2019) and inherits the second-order automation margin of Acemoglu and Restrepo (2018).

We illustrate the framework with two technologies run through the same economy: a broad field in the high-priced socio-cognitive north, calibrated to task-level AI exposure, and a narrow field on the low-priced manual arc, placed to represent industrial automation. Both release labour in their own automated region at plausible demand elasticities, while the wage effect is muted in the manual direction because depth is unpriced there. Read across elasticities, the model reproduces the contrast between inelastic-demand sectors that shed labour, such as mechanised agriculture, and elastic-demand sectors that grew employment under automation, such as nineteenth-century textiles, and places the consequences of cognitive automation as conditional on the elasticity of demand for cognitive output. The empirical realisation is illustrative: calibrated where task-level exposure data permit and placed by hand where they do not.

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1 Introduction

The division of labor segments the labor market into specialized domains. Workers accumulate human capital specific to their occupation or sector, and technologies are adopted unevenly across the task distribution. When a technology displaces workers from one set of tasks, the consequences for wages and employment depend on whether new tasks are created to absorb the displaced labor: the reinstatement mechanism of Acemoglu and Restrepo (2019). In the canonical one-dimensional framework, reinstatement unambiguously raises the labor share because new tasks are by construction assigned to labor at the boundary of the technology's operating domain (Acemoglu and Restrepo 2018). The question of which workers benefit is not well posed in a model where labor enters as two or three aggregate skill groups distinguished only by their position on a single task index.

This paper develops a spatial generalization of the task-based framework and tests its central prediction against occupational employment data. The framework places occupations as task bundles in a two-dimensional task space calibrated from semantic embeddings of O*NET task statements (Storck and Andersson 2026). The two dimensions capture domain orientation and specialization depth, placing occupations in a continuous space that reflects the actual content of work rather than predefined skill categories. The market price of skill is a field over this space, directional rather than uniform. Technologies are represented as augmentation fields: Gaussian productivity enhancements centered at a point in task space with a radius that determines their reach across occupational domains.

The spatial structure locates where new tasks appear and which of them survive as human work. Reinstatement seeds new tasks on the gradient boundary of the technology field, the ring at distance z_K from its centre where automated outputs must be validated, coordinated, and integrated. But a seeded task survives as *human* work only where labour is still the cheaper producer; deep in the core, where capital dominates, the same seed is captured by capital. The boundary ring is therefore where reinstatement is both created and, by the price of labour against capital, retained for workers. This derives, rather than assumes, the locus at which reinstatement accrues to labour in Acemoglu and Restrepo (2019).

Whether that reinstatement, and the automation that drives it, raises or lowers employment in a direction turns on the price of the work and the elasticity of demand for its output. Capital bears a task only where it is the cheaper producer, so it takes the dearest work first, and the per-unit cost saving it yields is second-order at the adoption margin: where capital just breaks even with labour the saving is zero, and it grows only deep in the core. Under competitive pricing the saving is passed to consumers as a lower price, and demand of elasticity η scales the revenue, and hence the labour demand, of the cheapened place. Elastic demand draws labour into the automated direction; inelastic demand releases it. Because the saving is second-order at the margin, displacement dominates at plausible elasticities, and net employment growth in an automated direction requires either elastic demand or automation deep enough to realise a large saving. This is the spatial form of the demand mechanism of Bessen (2019): the same labour-saving technology grows employment where demand is elastic and sheds it where demand is inelastic, and inherits the second-order automation margin of Acemoglu and Restrepo (2018).

We illustrate the framework with two technologies run through the same calibrated economy. The first is a broad field in the high-priced socio-cognitive north, with position and reach calibrated to task-level AI exposure mapped onto the geometry. The second is a narrow field on the low-priced manual arc, placed by hand to represent industrial automation, for which no task-level exposure measure on the modern classification exists. Both release labour in their own automated region at plausible demand elasticities, and the wage benefit of any reinstatement is muted in the manual direction because depth is unpriced there. Read across elasticities the two fields reproduce a known contrast: inelastic-demand mechanisation, as in agriculture, sheds labour, while the elastic-demand mechanisation of nineteenth-century textiles grew employment for decades (Bessen 2019). The consequences of cognitive automation are correspondingly conditional on how elastic the demand for cognitive output proves to be. We treat the empirical realisation as illustrative throughout, calibrated where exposure data permit and placed by hand where they do not, rather than as a causal test.

The paper proceeds as follows. Section 2 introduces the spatial occupational geometry. Section 3 presents the task-based production framework and its spatial extension, including the price of skill, the augmentation field, displacement, and displacement-anchored reinstatement through a bundle operator. Section 4 describes the calibration of technology parameters against empirical AI exposure data. Section 5 illustrates the model with two technologies, a calibrated cognitive field and a placed industrial field, and traces their employment and wage effects across the demand elasticity. Section 6 examines the consistency of the displacement channel with observed wage changes. Section 7 discusses implications and limitations.

2 The Spatial Occupational Geometry

[TODO: Kort presentation av geometrin från Paper 1: taskspace, vinkeldimension (domän), radiedimension (specialisering), occupational centroids och kompetensradier. En figur (taskspace-kartan). Ca 1–1.5 sidor.]

The value of this representation can be shown through its ability to illustrate successive waves of labor-replacing technology. The routine-task automation studied by Autor, Levy, and Murnane (2003) targeted occupations near the Technical-Physical pole, contributing to the secular decline of middle-wage manufacturing employment (Acemoglu and Restrepo 2020; Graetz and Michaels 2018; De Vries et al. 2020). Generative AI capabilities (Hampole et al. 2025; Eloundou et al. 2024) extend into parts of the Social-Cognitive region previously considered safe, raising concerns about the exposure of college-educated work. Each technology can be characterized by the region of the task space it affects, and each occupation by its position relative to that region. This is a question the geometry is built to address but that a partitioned or one-dimensional model cannot express.

3 Model

We embed the geometry into a production framework that builds on the canonical task-based model of Acemoglu and Restrepo (2019), extending it by replacing abstract factor categories with a continuous task space and by letting technology redraw the task bundles that define occupations.

3.1 Overview

The model has two layers. A *task layer* is a set of fields over the task disk. It carries production, the technology, displacement, reinstatement, and the creation of unbound tasks. A *worker layer* carries employment, mobility, and labor supply. The two layers are linked by the bundles that define occupations, and it is these bundles that automation and reinstatement rewrite.

The task layer is family-free. Job families are an external classification by industrial context, and the geometry of Storck and Andersson (2026) is built without such categories. We work at the level of tasks, the level at which the geometry is constructed and at which technology exposure is measured. Occupations enter as bundles of tasks, not as primitives.

Wages are pinned by the price of skill and the task bundles alone, and the theory layer carries no occupation-specific wage parameter. The price of skill is a field over the disk, identified by Storck and Andersson (2026) and carried here as the price of labor at each task location. What the field leaves unexplained is recorded in a measurement layer: a family-level wage wedge η_g, measured once on the cross-section and held fixed under technology shocks. The wedge is not part of the theory; simulations are reported with and without it (Section 4).

The model's objects accordingly divide into three layers with different epistemic status. The *theory* is two objects: the capability plane and the price functional (Section 3.2). The *measurement* is their frozen coordinates: the plane coefficients, the components of the price functional, and the wage wedge, each estimated once on the cross-section and never free under shocks. The *simulation* owns the free primitives: the technology (p_K, z_K, A_K, s_K) and the economy (R, τ, γ, ℓ, β, c, κ, η), where η is the elasticity of output demand introduced in Section 3.7. A technology scenario varies the last list only.

3.2 The Capability Plane and the Price of Skill

Storck and Andersson (2026) shows that the labor market does not value position in task space isotropically. The return to radial depth is positive toward the analytical directions and absent toward the service and manual directions, and what the market prices is the socio-cognitive cluster of capabilities concentrated in the northern half.

These findings are not independent, and the model rests on the structure that unifies them. Storck and Andersson (2026) establishes four regularities about how capabilities organize over the disk: the mean similarity of capability deviation profiles between occupations follows $a \cos \Delta\xi$; the distance of a profile from the mean profile grows with χ while its shape concentration does not; the wage return to χ has no isotropic component; and the directional return is a single first harmonic. One structure generates all four. We posit that the capability content of work is, to first order, a two-dimensional plane in capability space,

$$v_o \;=\; \bar{v} \;+\; \chi_o\Big(\cos\xi_o\, u_1 + \sin\xi_o\, u_2\Big) \;+\; \varepsilon_o, \tag{1}$$

where v_o is the occupation's capability profile, $\bar{v}$ the mean profile, and $u_1 \perp u_2$ the two poles of the plane. The task disk is the polar map of this plane: χ is amplitude, ξ is phase. Under the plane, the cosine law for deviation similarity is exact, with noise attenuating it to $a \cos \Delta\xi$ and more so for central pairs; the deviation magnitude is χ itself; the profile shape is free of χ; and distance on the disk is, up to the common pole norm, distance in capability space. This is the structural postulate of the model. Every field the model carries over the disk, requirements and prices alike, inherits its functional form from it; higher-order structure is texture, measured and reported in Section 4 but carried outside the theory. Section 4 also tests the postulate directly: the leading two components hold roughly three quarters of the capability deviation variance, with a sharp gap to the third, and the leading score plane aligns with the disk coordinates.

The market enters through one further primitive: capabilities are priced by a linear functional, so the log price of a unit of work is affine in the capability content of its location, $\ln \Pi(\mathbf{r}) = \kappa + \sum_k p_k\, q_k(\mathbf{r})$. Under the plane this implies that the depth return is a single first harmonic, $\chi\,(p{\cdot}u_1 \cos\xi + p{\cdot}u_2 \sin\xi)$, and that the isotropic depth term is zero; both are confirmed in the estimation (Section 4). We summarize the price of skill as a field $\Pi(\mathbf{r})$ over the disk, the price the market pays for a unit of work performed at task location $\mathbf{r} = (\xi, \chi)$. It is the reduced form of the directional wage equation of Storck and Andersson (2026), a first-harmonic field on the circle whose radial term is modulated by direction,

$$\ln \Pi(\mathbf{r}) \;=\; m_0 + m_1 \cos\xi + m_2 \sin\xi \;+\; \chi\,\Big(m_3 + m_4 \cos\xi + m_5 \sin\xi\Big). \tag{2}$$

The direction-modulated radial term is what the plane generates, and it is what makes depth rewarded in the north and unrewarded in the south. The level terms m_1, m_2 are measured structure outside the plane: the part of the wage that does not run through depth-built capabilities, retained in the reduced form and quantified as texture in Section 4. Π is estimated once, on the cross-section of occupations, and treated as fixed structure here. Calling it a reduced form is precise: the coefficients are identified at occupational centroids, where wages are observed, while the model reads the field at task locations, where production happens. The step from the one to the other is testable, because the model prices an occupation as its bundle, eq. (3) below, not as the field at its centroid; Section 4 shows that the bundle-integrated wage reproduces observed wages as well as the centroid evaluation does, and quantifies the gap between the two.

The field is identified on the occupational support, which extends to $\chi = 0.752$, the largest centroid radius in the estimation sample; tasks extend to $\chi = 1$. We treat (2) as structure over the full disk, so evaluating it at peripheral task locations is a linear extrapolation in χ beyond the occupational support. The task layer is defined to the rim, and the field follows it; Section 4 reports the quantitative weight of the extrapolated region.

Productivity is isotropic. Storck and Andersson (2026) does not find that workers are more productive in the north. It finds that the market pays more for the capabilities concentrated there. Productivity belongs to the worker and is symmetric; the price of that work is directional. The directional structure therefore lives in Π, not in a productivity kernel.

3.3 Tasks, Occupations, and Wages

A task is a unit of work activity that yields a unit of output. Because one task is one unit of output, no productivity magnitude varies across tasks. What varies over the disk is how much labor is present and what that labor is paid.

An occupation o is a bundle: a weight measure $b_o(\mathbf{r})$ over task space, normalized to one, weighted by the O*NET task-relevance ratings, the same weights that define the occupational centroids in Storck and Andersson (2026). Its centroid $\mu_o = \int \mathbf{r}\, b_o(\mathbf{r})\, d\mathbf{r}$ is a summary of the bundle, not the object itself. The occupation's wage is the price of its bundle,

$$w_o \;=\; \int \Pi(\mathbf{r})\, b_o(\mathbf{r})\, d\mathbf{r}, \tag{3}$$

the average price of skill over the tasks the occupation performs. The wage is therefore not a property of the occupation but an aggregate of the price field over its current bundle, and it moves when the bundle is rewritten. Labor density on the disk is

$$n(\mathbf{r}) \;=\; \sum_o L_o\, b_o(\mathbf{r}), \tag{4}$$

where L_o is employment in occupation o, the employment-weighted task coverage of the disk.

3.4 Technology and the Operated Regime

A technology K is a field $\phi_K(\mathbf{r})$, a Gaussian centered at p_K with reach z_K,

$$\phi_K(\mathbf{r}) \;=\; A_K \exp\left(-\frac{1}{2}\left(\frac{\|\mathbf{r}-p_K\|}{z_K}\right)^2\right), \tag{5}$$

the effectiveness with which capital can do the work at $\mathbf{r}$. Capital does not perform a task autonomously. At a location the worker remains and the output is unchanged; what the technology changes is the share of the location's micro-tasks that capital bears. The position p_K fixes where in task space the technology bites, and the reach z_K fixes how far it extends: a narrow z_K is a specialized technology, a broad z_K a general-purpose one. Producing the crossed micro-tasks more cheaply with capital is the productivity effect of automation in the sense of Acemoglu and Restrepo (2018), here made spatial.

A technology also has a character. The same effectiveness can replace the worker or assist her, and these have different consequences. We carry the character as a scalar $s_K \in [0,1]$, a fourth attribute of the technology beside p_K, z_K, and A_K. At $s_K = 1$ the technology is purely replacing, at $s_K = 0$ purely augmenting. The effectiveness splits into a replacing part $s_K\,\phi_K$ and an augmenting part $(1-s_K)\,\phi_K$, which act on the same location through different channels. The augmenting part assists rather than replaces; we treat it in Section 3.9.

Whether capital bears a micro-task is a question of price, and only the replacing part competes with the worker on it. Producing the task with labor costs $\Pi(\mathbf{r})$, the price of the work. Producing it with the replacing part of capital costs the rental R against that part's effectiveness, an effective cost $R/(s_K\,\phi_K(\mathbf{r}))$. Capital is the cheaper producer when

$$s_K\,\phi_K(\mathbf{r}) \;>\; \frac{R}{\Pi(\mathbf{r})}. \tag{6}$$

The comparison is between the rental of capital and the price of labor at the location, weighted by the technology's effectiveness there. The threshold $R/\Pi(\mathbf{r})$ is lowest where $\Pi(\mathbf{r})$ is highest, so capital is profitable first at the most highly priced locations, and a broad, inexpensive technology can sustain it well beyond its core. Capital moves first against the dear work. This is the spatial counterpart of the threshold in Acemoglu and Restrepo (2018): where they automate the lowest task on a single scale, the field takes the highest-priced positions wherever they lie on the disk.

A location $\mathbf{r}$ is not a single task. It is a neighborhood holding a continuum of micro-tasks, and the effectiveness ϕ_K varies across them, so (6) is not true or false for the whole cell at once. It holds for a fraction of the micro-tasks and fails for the rest. We write that fraction as $a(\mathbf{r})$, the degree to which capital bears the work at $\mathbf{r}$,

$$a(\mathbf{r}) \;=\; \frac{1}{1+\exp\!\Big(-(s_K\,\phi_K(\mathbf{r}) - R/\Pi(\mathbf{r}))/\tau\Big)}, \tag{7}$$

where τ sets the width of the margin, the dispersion of (6) across the micro-tasks of the cell. As $\tau \to 0$ the cell collapses to a single task and $a(\mathbf{r})$ becomes a step, zero below the threshold and one above it; in that limit the operated region is a sharp partition of task space and the margin is a boundary in space. For $\tau > 0$ the margin lies inside each cell, and $a(\mathbf{r})$ is the share of its micro-tasks on the operated side of (6).

The replacing part sets the takeover; the augmenting part does not enter it. At $s_K = 1$ the threshold (6) and the operated share (7) are those of pure replacement, and the displacement, wage, and density results that follow hold unchanged. The replacement model is the case $s_K = 1$.

Displacement is then a loss of priced contribution, not a loss of the task. The worker remains at $\mathbf{r}$ and the output is unchanged; what shrinks is the share of that output the worker is paid for. The priced contribution of the cell is $\Pi(\mathbf{r})$ scaled by the retained share $1 - a(\mathbf{r})$. The mass displaced from occupation o is the operated share integrated over its own bundle,

$$D_o \;=\; \int b_o(\mathbf{r})\,a(\mathbf{r})\,d\mathbf{r}, \tag{8}$$

and its retained wage is $\int \Pi(\mathbf{r})\,b_o(\mathbf{r})\,(1 - a(\mathbf{r}))\,d\mathbf{r}$. The aggregate displaced labor mass is $\Delta\Gamma^D = \sum_o L_o D_o$. An occupation can carry substantial D_o without any region having become operated as a whole, because the displacement is spread across its bundle rather than concentrated in a zone it owns. This is the difference between an occupation disappearing and an occupation that remains but is paid for less of what it does.

The takeover also lowers the cost of producing the cell. Where capital bears the share $a(\mathbf{r})$ it does so at the effective rental $R/(s_K\,\phi_K(\mathbf{r}))$ rather than the price of labour $\Pi(\mathbf{r})$, so the unit cost of the work at $\mathbf{r}$ falls from $\Pi(\mathbf{r})$ to

$$c(\mathbf{r}) \;=\; \Big(1 - a(\mathbf{r})\Big)\,\Pi(\mathbf{r}) \;+\; a(\mathbf{r})\,\frac{R}{s_K\,\phi_K(\mathbf{r})}, \tag{9}$$

the labour-weighted price plus the capital-weighted rental. The realised per-unit saving is $\Pi - c = a\,(\Pi - R/(s_K\phi_K))$, positive wherever capital is adopted. Two features of this saving carry the model's central comparative statics. It is largest deep in the technology's core, where the price gate (6) is crossed by the widest margin, and it is *second-order at the adoption margin*: as $a \to 1/2$ the capital cost $R/(s_K\phi_K)$ approaches Π, so $c \to \Pi$ and the saving vanishes. The marginal task that capital just takes is one it produces at the price labour charged, the spatial form of the second-order automation margin of Acemoglu and

Restrepo (2018). Where this saving goes, and what it does to employment, is settled by the output market in Section 3.7.

Because $a(\mathbf{r})$ varies within the bundle, the margin cuts inside the occupation rather than between regions. Some of an occupation's own tasks are borne by capital while others remain with the worker. This is the restructuring of work within an occupation, the unbundling and rebundling of Brynjolfsson, Mitchell, and Rock (2018), expressed as a field. The reach of the technology, p_K and z_K, together with the price field Π, determines which tasks within each bundle cross the margin.

3.5 Reinstatement: The Bundle Operator

Reinstatement returns a fraction γ of the displaced mass as new tasks, and the new tasks must attach to occupations. Free tasks have no role in the model. We therefore need a rule for where new tasks appear and which occupations can take them.

Where they appear. The gradient of the augmentation field,

$$\|\nabla\phi_K(\mathbf{r})\| \;=\; \frac{1}{z_K^2}\,\|\mathbf{r}-p_K\|\cdot\phi_K(\mathbf{r}), \tag{10}$$

peaks at distance z_K from p_K. New tasks are seeded on this boundary ring. With $\hat{g}(\mathbf{r}) = \|\nabla\phi_K(\mathbf{r})\|/\int\|\nabla\phi_K\|$ the normalized ring density and $M = \gamma\,\Delta\Gamma^D$ the seeded mass, the seeded density is $s(\mathbf{r}) = M\,\hat{g}(\mathbf{r})$.

Which seeds survive as human work. The ring density $\hat{g}$ records where new tasks are *seeded*, the potential for task creation, not where they survive as work for labour. A seed at $\mathbf{r}$ competes with capital on the same terms as any task there: it survives as human work only to the extent that labour is the cheaper producer, the retained share $1-a(\mathbf{r})$. Deep in the core, where $a \to 1$, a seeded task is taken by capital as readily as the tasks already present; on the ring, where the field has fallen back and a is small, it survives with the worker. The surviving human seed density is therefore $s(\mathbf{r})\,(1-a(\mathbf{r}))$, and the residual $s(\mathbf{r})\,a(\mathbf{r})$ is captured by capital. This is where the model earns, rather than assumes, the result that reinstatement accrues to labour: Acemoglu and Restrepo (2019) place new tasks where labour holds the comparative advantage by construction; here the price gate $1-a(\mathbf{r})$ selects them out of a ring that is seeded irrespective of who will bear the work.

Which occupations can take them. A new task at $\mathbf{r}$ requires a profile of capabilities. The requirement fields are not a modeling choice: under the postulate of Section 3.2 they are the plane itself, read per capability cluster,

$$q_k(\xi,\chi) \;=\; \bar{v}_k \;+\; \chi\left(u_{1,k}\cos\xi + u_{2,k}\sin\xi\right), \tag{11}$$

for $k \in \{\mathrm{S1}, \mathrm{S2}, \mathrm{A1}, \mathrm{A2}\}$, with coefficients fitted once to occupation-level cluster intensities (Section 4); a measured variant that adds the level harmonics and an isotropic depth term is carried as a sensitivity. The fields read the requirement at a location from the capabilities of the occupations positioned there, a revealed-requirement assumption. An occupation can attach the task readily when it already holds those capabilities, and only with difficulty when the task requires a capability the occupation lacks. The relevant quantity is the deficit, the positive shortfall, not a symmetric distance:

$$\delta_o(\mathbf{r}) \;=\; \sum_{k \in \mathcal{K}} v_k \, \max\Big(q_k(\mathbf{r}) - q_{o,k},\, 0\Big), \tag{12}$$

where $q_k(\mathbf{r})$ is the level of capability cluster k required at $\mathbf{r}$, $q_{o,k}$ is the occupation's level on cluster k, v_k is the weight of cluster k, and $\mathcal{K}$ is the set of priced clusters. Attaching tasks that use less of the occupation's capabilities is free. Attaching tasks that demand more, or a different kind, is costly.

The weights are not free parameters: they are the components of the price functional of Section 3.2, $v_k = p_k$, measured by the mediation analysis of Storck and Andersson (2026) and replicated on the frozen inputs (Section 4): large for the socio-cognitive skill cluster S1, small for the technical cluster S2, not significant for the ability clusters A1 and A2. The restriction to priced clusters follows from the same identification, not from a significance convention: a capability the functional does not price lies in its kernel, and a shortfall in it is no economic barrier in the sense the operator formalizes; the sign restriction $v_k \geq 0$ excludes carrying noise estimates as theory. Empirically $\mathcal{K} = \{\mathrm{S1}, \mathrm{S2}\}$; the all-cluster variant is a sensitivity. A single functional thus does two jobs. It prices the work a location demands, and it gates what a bundle can absorb. New tasks attach freely only where the deficit is small, that is, where the priced capability content does not exceed what the occupation already holds. Reinstatement can therefore broaden a bundle but cannot lift its priced content. The wage cannot jump. A large jump would require a capability the occupation does not have, and that is not reinstatement within an occupation but the seed of a new one.

In the one-dimensional framework, comparative advantage is a monotone schedule: labor is strongest in the highest-indexed tasks, and new tasks are added there by construction (Assumption 1 of Acemoglu and Restrepo (2018)). The deficit gate is the geometric counterpart. It does not posit a single direction in which labor is strong. It lets a new task attach where the occupation already holds the priced capability the task requires, wherever that lies on the disk.

Attachment and allocation. The occupation's readiness to attach a task at $\mathbf{r}$ is

$$e_o(\mathbf{r}) \;=\; \exp\Big(-\delta_o(\mathbf{r})/\ell\Big), \tag{13}$$

between zero and one, where ℓ is a single global scale. The total attachment capacity at $\mathbf{r}$ is $C(\mathbf{r}) = \sum_o L_o\, e_o(\mathbf{r})$, the labor present weighted by its readiness. A saturating function $\Phi(C) = C/(1 + C)$ gives the bound fraction of the surviving seed mass. Occupation o receives the reinstated density

$$\iota_o(\mathbf{r}) \;=\; s(\mathbf{r})\left(1 - a(\mathbf{r})\right)\Phi(C(\mathbf{r}))\,\frac{L_o\, e_o(\mathbf{r})}{C(\mathbf{r})}, \tag{14}$$

its share of the bound surviving mass in proportion to its readiness. The price gate $1 - a(\mathbf{r})$ and the deficit gate $e_o(\mathbf{r})$ act in series: the first decides how much of the seed survives capital at $\mathbf{r}$, the second how that surviving mass is shared among the occupations that can hold it.

The bundle operator. The post-technology bundle removes displaced weight and adds reinstated weight,

$$b_o^{\text{post}}(\mathbf{r}) \;=\; b_o(\mathbf{r})\left(1 - a(\mathbf{r})\right) \;+\; \frac{\iota_o(\mathbf{r})}{L_o}. \tag{15}$$

The same field ϕ_K drives both terms. It deactivates through $a(\mathbf{r})$, high near the core, and seeds through $\hat{g}(\mathbf{r})$, high on the ring. The bundle is rewritten continuously as the technology intensifies. This is the geometric form of occupational redefinition (Brynjolfsson, Mitchell, and Rock 2018): the occupation does not vanish, its bundle is redrawn.

The operator does not preserve the bundle's mass. Integrating (15) gives $\int b_o^{\text{post}} = 1 - D_o + B_o/L_o$, where $B_o = \int \iota_o\, d\mathbf{r}$ is the seeded mass the occupation binds. We do not renormalize. Renormalization would assert that the freed time is spread proportionally over the remaining tasks, a behavioral claim that presupposes slack demand for additional human work at the retained locations, which the place output of Section 3.7 does not provide. It would also double-count the model's actual channel for freed labor, the re-sorting of the worker layer. The mass deficit is that channel's input: $D_o - B_o/L_o$ is, per worker, exactly the labor the occupation releases to the worker layer. After the operator, the bundle is a measure of performed and paid content rather than a unit weight, and the wage (3) is the priced content of that measure, not an average over it; the two readings coincide only pre-technology.

Wages and the Turing trap. The wage change is the price-weighted change in the bundle,

$$\Delta w_o \;=\; \int \Pi(\mathbf{r})\left[b_o^{\text{post}}(\mathbf{r}) - b_o(\mathbf{r})\right] d\mathbf{r} \;=\; \int \Pi(\mathbf{r})\left[-b_o(\mathbf{r})\, a(\mathbf{r}) + \tfrac{\iota_o(\mathbf{r})}{L_o}\right] d\mathbf{r}. \tag{16}$$

The first term is negative and falls on high-Π tasks; the second adds tasks that, by the deficit gate, are mostly low-Π. Default reinstatement therefore carries a downward wage bias: it strips priced work and refills with unpriced work. The sign of Δw_o is the contest between where $a(\mathbf{r})$ removes weight and where ι_o adds it, read against the price gradient $\nabla\Pi$. In the directions where depth is priced, the strip falls on high-Π tasks and the refill cannot replace them.

Density. Summing the bundle change over occupations gives the evolved task density,

$$\rho_1(\mathbf{r}) - \rho_0(\mathbf{r}) \;=\; -\sum_o L_o\, b_o(\mathbf{r})\, a(\mathbf{r}) \;+\; \sum_o \iota_o(\mathbf{r}) \;+\; u(\mathbf{r}), \tag{17}$$

with $u(\mathbf{r})$ defined in Section 3.6. The seeded mass integrates to $\gamma\,\Delta\Gamma^D$, so γ keeps its interpretation as the fraction of displaced mass returned as new tasks; of that seed the human-surviving share $\int s(1-a)$ is added as task-work and the remainder $\int s\,a$ is captured by capital. Setting $\gamma = 0$ recovers pure automation; $\gamma = 1$ seeds the full displaced mass, of which the price gate still decides how much survives as human work.

3.6 Unbound Tasks and Candidate Occupations

Not all surviving seed attaches. Where every occupation has a large deficit, $C(\mathbf{r})$ is small, $\Phi(C(\mathbf{r}))$ is near zero, and the surviving human seed stays unbound:

$$u(\mathbf{r}) \;=\; s(\mathbf{r})\,\Big(1 - a(\mathbf{r})\Big)\,\Big[1 - \Phi(C(\mathbf{r}))\Big]. \tag{18}$$

u is the density of new tasks that survive capital but that no existing occupation can perform. It is an output of the model, not an agent. Each seeded location has three fates that partition its mass: the share $a(\mathbf{r})$ captured by capital, the bound human share $(1-a)\Phi$ absorbed by existing occupations, and the unbound human share $(1-a)(1-\Phi)$. These integrate to the seeded $\gamma\,\Delta\Gamma^D$, so the closed accounting holds across all three. In the static model this unbound share is created and read off at a single equilibrium; developing it as a stock that fills as the technology matures and drains as occupations bind it, so that its trajectory measures the transitional mismatch between displacement and reinstatement, is the subject of ongoing work.

The map of u answers two distinct questions. Where unbound tasks are created is u itself, the ring regions that are active but that no bundle can absorb. Where new occupations may emerge is where u concentrates, a connected region carrying enough mass to support an occupation, not merely a high value at a point. A diffuse u along the ring is friction. A clump is a candidate.

The price field tells the two apart. Unbound mass in a high-Π region is latent skilled work awaiting a bearer, the kind of region from which qualified new occupations have historically grown. Unbound mass in a low-Π region is work no occupation wants and no worker is paid well for, closer to precarious friction than to a new occupation. We color candidate regions by Π and read opportunity against spillage. We do not instantiate new occupations: the model shows where one may appear, not that one does.

The same deficit that gates attachment draws the boundary between two literatures. New tasks are absorbed by existing occupations where the priced deficit is small, which is redefinition in the sense of Brynjolfsson, Mitchell, and Rock (2018). They remain unbound where the deficit is large for every occupation, which is the new work of Acemoglu and Restrepo (2019) that no existing occupation owns. One operator decides which, through the deficit.

3.7 Employment and Mobility

The worker layer sets employment L_o and closes the model. We solve it as a static spatial-sorting equilibrium and read the effect of a technology as the comparative static between two such equilibria, one with the original bundles and one with the bundles rewritten by (15). A law of motion in time can be added on top of this rest point; we do not need it for a statement about the new equilibrium.

Two factors share each place. A location is produced by human and capital input as perfect substitutes, in the manner of Acemoglu and Restrepo (2018). The bundle operator splits each occupation's coverage of a place into a human part $h_o(\mathbf{r}) = b_o^{\mathrm{post}}(\mathbf{r})$, the work the worker performs, and a capital part $k_o(\mathbf{r}) = b_o(\mathbf{r})\, a(\mathbf{r})$, the work capital bears. Total task-work at $\mathbf{r}$ is

$$n(\mathbf{r}) \;=\; \sum_o L_o \Big(h_o(\mathbf{r}) + k_o(\mathbf{r})\Big) \;=\; \sum_o L_o\, b_o(\mathbf{r}) \;+\; \sum_o \iota_o(\mathbf{r}), \tag{19}$$

the pre-technology density (4) plus the bound seeded mass; unbound mass performs no work and does not enter. The operated share enters once, inside h_o; no further $(1-a)$ factor is applied downstream. Pre-technology, $h_o = b_o$, $k_o = 0$, and n reduces to (4). Output is sold, and the demand for it responds to its price. Under competitive pricing the unit price of the work at $\mathbf{r}$ is its marginal cost $c(\mathbf{r})$ of eq. (9), so the automation saving $\Pi - c$ is passed to buyers as a lower price. With isoelastic demand of elasticity η, the quantity bought scales as $(\Pi/c)^\eta$ and the revenue of the place as $c\,(\Pi/c)^\eta = \Pi\,(c/\Pi)^{1-\eta}$. We carry this as a demand multiplier on the price field,

$$D(\mathbf{r}) \;=\; \left(\frac{c(\mathbf{r})}{\Pi(\mathbf{r})}\right)^{1-\eta}, \qquad V(\mathbf{r}) \;=\; D(\mathbf{r})\,\Pi(\mathbf{r})\,n(\mathbf{r})^{\beta}, \qquad \beta \in (0,1), \tag{20}$$

concave in task-work through the congestion exponent β. The multiplier is the model's one channel from the wider economy, and the elasticity η its only new primitive; the closure remains partial, in that each place faces an independent isoelastic demand with no cross-place substitution or income effects. At $\eta = 1$, unit-elastic demand, $D \equiv 1$: the cost saving leaves revenue unchanged and the place output is the cost-invariant $\Pi\, n^{\beta}$ one would write if output value were held fixed. For $\eta > 1$, elastic demand, $D > 1$ where cost has fallen, so the cheapened direction commands more revenue and draws labour toward it; for $\eta < 1$, inelastic demand, $D < 1$ there, and labour is released. Because the saving $\Pi - c$ is second-order at the adoption margin (Section 3.4), D departs from one only where automation runs deep, so the demand channel is weak at the margin and the displacement of the takeover dominates unless demand is strongly elastic. Congestion acts on n, the total task-work present, regardless of which factor bears it; the takeover redistributes the work between h_o and k_o without changing their sum at a location. Because the pre-technology bundle b_o is normalized, a broad task radius spreads it thin and contributes little to $n(\mathbf{r})$, so a concentrated occupation presses harder on the places it occupies than a diffuse one does.

Returns to the two factors. Human and capital are paid the marginal product of task-work, $\beta\, D(\mathbf{r})\, \Pi(\mathbf{r})\, n(\mathbf{r})^{\beta-1}$, each on the part of the work it performs: the human side on $\sum_o L_o\, h_o(\mathbf{r})$, the capital side on $\sum_o L_o\, k_o(\mathbf{r})$. Their payments sum to $\beta V(\mathbf{r})$; the residual $(1-\beta)V(\mathbf{r})$ is the congestion rent to the fixed local factor.

Occupation value and equilibrium. A worker carries the whole bundle and earns the marginal product integrated over the content she performs,

$$W_o \;=\; \beta \int h_o(\mathbf{r})\, D(\mathbf{r})\, \Pi(\mathbf{r})\, n(\mathbf{r})^{\beta-1}\, d\mathbf{r}, \tag{21}$$

the congestion-, takeover-, and demand-adjusted counterpart of the gross bundle price w_o of (3): the takeover enters through the stripped part of h_o, congestion through $n^{\beta-1}$, the price through Π, and the output market through D. Occupation values are coupled, because a location's density sums over every occupation that covers it, so occupations that overlap in task space press on one another's returns. Labor re-sorts from the pre-technology distribution L^0_o, drawn toward value and damped by the bundle distance $d(o,o') = \|\mu_o - \mu_{o'}\|$, the Euclidean distance between occupational centroids, which Storck and Andersson (2026) ties to the transferability of human capital in the sense of Gathmann and Schönberg (2010). Under the capability plane the metric is induced rather than chosen: with orthogonal poles of equal norm, distance on the disk is proportional to distance in capability space, so the damping measures the capability gap a mover must close. The allocation is a logit over value net of the mobility cost from origin,

$$L_o \;=\; L^{\text{tot}}\, \frac{L^0_o\, \exp\!\Big((W_o - c\, d(o, o^0))/\kappa\Big)}{\sum_{o'} L^0_{o'}\, \exp\!\Big((W_{o'} - c\, d(o', o^0))/\kappa\Big)}, \tag{22}$$

a fixed point because W_o depends on $n(\mathbf{r})$, which depends on $\{L_o\}$. The distance term bites in a static solution because the re-sorting is measured from an origin, not from nowhere. Total labor is fixed at L^{tot}; letting it respond to aggregate value is a later extension.

The labor share. With $H(\mathbf{r}) = \sum_o L_o\, h_o(\mathbf{r})$ the human task-work, total labor payment is $\int \beta\, D(\mathbf{r})\, \Pi(\mathbf{r})\, H(\mathbf{r})\, n(\mathbf{r})^{\beta-1}\, d\mathbf{r}$ and total factor payment to human and capital is $\int \beta\, D(\mathbf{r})\, \Pi(\mathbf{r})\, n(\mathbf{r})^{\beta}\, d\mathbf{r}$, so the labor share of factor income is

$$\Lambda \;=\; \frac{\int D(\mathbf{r})\, \Pi(\mathbf{r})\, H(\mathbf{r})\, n(\mathbf{r})^{\beta-1}\, d\mathbf{r}}{\int D(\mathbf{r})\, \Pi(\mathbf{r})\, n(\mathbf{r})^{\beta}\, d\mathbf{r}}. \tag{23}$$

The congestion exponent β cancels: it governs the spatial sort but not the share. The demand multiplier D does not cancel, since it varies across places, so demand reweights the share toward the directions whose output it expands; at $\eta = 1$ it is one and the share is the price- and density-weighted mean of the human share $H(\mathbf{r})/n(\mathbf{r})$, which is one pre-technology; automation lowers it by raising a where (6) is crossed, and with no reinstatement it reduces to the mean of $1 - a(\mathbf{r})$. Solving (22) once with the original bundles and once with the rewritten bundles gives two equilibria, and the difference in Λ is the effect of the technology. The displacement of labor onto fewer tasks then appears not at the automated places, where a is high and density falls, but at their destinations, where the re-sorting raises n and congestion bites. The bunching of Acemoglu and Restrepo (2018) is recovered as an equilibrium effect of the sort rather than assumed locally, and whether the share falls hardest in priced or unpriced directions is decided by where a rises against the price field.

3.8 Augmentation and the Character of the Technology

3.9 The Character of the Technology and the Incidence of the Saving

The character $s_K \in [0, 1]$ is the share of the field's effectiveness that competes with labour at the takeover gate: it enters the operated share (7) through $s_K\, \phi_K$ and nowhere else. At $s_K = 1$ the whole field replaces; below one, only the fraction s_K of it bears on the price comparison (6), and the takeover is correspondingly smaller. The main model takes $s_K = 1$. The model carries no separate augmenting term that lowers the labour a worker needs at unchanged output; the productivity gain of the technology is the cost saving $\Pi - c$ of Section 3.4, and its consequence for labour is settled by the output market of Section 3.7, not by a second technology attribute.

This is where the augmentation–substitution distinction of Brynjolfsson, Mitchell, and Rock (2018) enters, as an outcome rather than an assumption. Who captures the saving is not posited. Under competitive pricing the firm earns no rent, the worker is paid the going price of labour Π on the tasks she still performs, and the saving flows to consumers as a lower price. Its employment consequence runs through the demand multiplier D: where demand is elastic the saving returns as labour demand in the cheapened direction and the technology reads as augmenting, raising employment; where demand is inelastic it does not, and the technology reads as substituting, releasing labour. The same physical technology is augmenting or substituting depending on the elasticity of demand for what it cheapens, which is why the character of automation cannot be read from the technology alone.

Replacement carries its own brake. As priced mass passes to capital the location's retained price falls, the threshold R/Π rises, and further takeover slows, the self-correction of Acemoglu and Restrepo (2018) inherited here through the price field rather than assumed. The labour freed by the takeover does not leave the model: it enters the mobile pool and re-sorts through the allocation (22), drawn toward value and damped by mobility cost, so where it lands is set by the priced locations it can reach and not by the location it left.

One bound on this treatment. The character s_K is a scalar, uniform over the reach of the technology; a technology that replaces in one direction and assists in another is a direction-dependent character $s_K(\mathbf{r})$, which we leave as an extension.

4 Empirical Implementation and Calibration

This section measures the frozen structure of the model: the capability plane, the price field, the bundle pricing it implies, the components of the price functional, and the wage wedge. Every number reported here is the output of a script in the replication repository, and the inputs are frozen copies of the Paper 1 exports with recorded provenance. The technology parameters, the only objects a scenario is allowed to vary, are calibrated in Section 4.7.

4.1 Data and frozen inputs

All inputs are vendored from a pinned commit of the Storck and Andersson (2026) repository by a single freezing script, which is the model's only contact surface with the upstream project; every analysis script reads exclusively from the frozen copies, and a manifest records the source commit, the encoder run, and per-file hashes and derivation recipes. The frozen inputs are the occupation and task polar coordinates of the reference encoder run (878 occupations; task coordinates and relevance ratings defining the bundles b_o), BLS OEWS May 2023 median hourly wages, the frequency-weighted mean required level of education per occupation, the capability cluster intensities $q_{o,k}$, and the occupation-by-descriptor matrices of raw levels for the 35 O*NET Skills and 52 Abilities. Merging coordinates with positive median wages and education reproduces the Paper 1 estimation sample of $N = 785$ occupations; the capability-side estimations, which do not require wages, use all 878.

4.2 The capability plane

The postulate of Section 3.2 is testable directly, and the test uses data that enter no step in the construction of the coordinates: the disk is built from task text alone, while the descriptor matrices record surveyed capability levels. For each descriptor family we compute the principal components of the deviations $v_o - \bar{v}$ and ask two questions: how much of the deviation variance lies in a two-dimensional structure, and whether that structure is the disk.

The answers are, respectively, three quarters and yes to first order. For Skills, the leading two components hold 73.3 percent of the deviation variance (46.2 and 27.0), with a sharp gap to the third (5.7); for Abilities, 75.4 percent (62.7 and 12.7, against 5.8 for the third). The leading score plane aligns with the disk coordinates $(x, y) = (\chi \cos \xi, \chi \sin \xi)$ with canonical correlations 0.93 and 0.72 for Skills and 0.89 and 0.56 for Abilities; the disk coordinates are recovered from the scores with $R^2 = 0.85$ (x) and 0.55 (y) for Skills, 0.64 and 0.49 for Abilities. The capability content of work is two-dimensional to first order, and the task disk is an imperfect but unmistakable polar map of that plane: the first canonical direction is nearly identified, the second is recovered more coarsely.

The plane read per capability cluster gives the requirement fields of eq. (11), fitted by OLS on the 878 occupations (Table 1). The pole structure matches the compass of Storck and Andersson (2026): the socio-cognitive clusters S1 and A1 point northeast (52° and 54°), the technical cluster S2 northwest (151°), the physical cluster A2 west (189°). The exact plane form, three parameters per cluster, fits nearly as well as the measured six-parameter variant: adding the level harmonics and the isotropic depth term raises R^2 by at most 0.009 (Table 1, last column). The theory costs almost nothing relative to the unrestricted first-harmonic fit.

Table 1: The capability plane read per cluster, eq. (11): OLS on $N = 878$ occupations, HC3 standard errors in parentheses. Pole direction is $\operatorname{atan2}(u_{2,k}, u_{1,k})$, amplitude $\|(u_{1,k}, u_{2,k})\|$. ΔR^2 is the gain from the measured variant (adding level harmonics and an isotropic depth term).

Cluster	$\bar{v}_k$	$u_{1,k}$	$u_{2,k}$	Pole	R^2	ΔR^2 (meas.)
S1	2.682 (0.011)	+1.094 (0.027)	+1.421 (0.048)	52°	0.737	+0.001
S2	1.366 (0.018)	−2.219 (0.048)	+1.228 (0.080)	151°	0.686	+0.008
A1	3.198 (0.009)	+0.952 (0.024)	+1.309 (0.041)	54°	0.748	+0.000
A2	1.573 (0.016)	−1.539 (0.038)	−0.244 (0.065)	189°	0.571	+0.009

What the plane does not generate is recorded in a texture ledger (Table 2), with the same discipline as the wage wedge: measured once, carried outside the theory, switched on in sensitivity runs. The entries are the level harmonics of the price field (the eastward gradient $m_1 = +0.19$), the east–west level second harmonic of wages (± 9 percent at fixed χ), the cluster-level gains of Table 1, and the second-harmonic remainders of the requirement fields, which are statistically detectable at $N = 878$ but add at most 0.030 (0.011) of R^2 in level (interaction).

Table 2: Texture ledger: measured structure outside the capability plane. Each entry is estimated once and held fixed; regime runs are reported with and without the texture-bearing variants.

Texture	Magnitude	Where measured
Wage level gradient (east)	$m_1 = +0.192$, dir. 25°	price-field estimation
Wage level 2nd harmonic	$\pm 9\%$ E–W at fixed χ	price-field estimation
q_k level harmonics + isotropic depth	$\Delta R^2 \leq 0.009$	plane estimation
q_k 2nd harmonics (level; interaction)	$\Delta R^2 \leq 0.030$; ≤ 0.011	plane estimation

4.3 The price field

The coefficients of eq. (2) are estimated by OLS of log median hourly wage on the polar regressors over the $N = 785$ sample, with HC3 standard errors. The estimation first replicates the geometry-only regression of Storck and Andersson (2026) exactly ($\cos\xi$ $+0.212$, $\sin\xi$ $+0.389$, χ -0.046, $R^2 = 0.492$), pinning the sample, and then adds the interaction terms. The estimated field is

$$\ln \Pi = \underset{(0.036)}{3.420} + \underset{(0.048)}{0.192} \cos\xi + \underset{(0.051)}{0.089} \sin\xi + \chi\Big(\underset{(0.090)}{-0.065} + \underset{(0.112)}{0.008} \cos\xi + \underset{(0.134)}{0.891} \sin\xi\Big),$$

$R^2 = 0.523$. The two predictions of the plane-with-linear-pricing structure hold: the isotropic depth term is indistinguishable from zero ($m_3 = -0.065$, $p = 0.47$), and the depth return is a single first harmonic, peaking at 89.5° with amplitude 0.891 and positive for $\xi \in (3.6°, 175.3°)$. The balanced second-harmonic test leaves the interaction terms jointly insignificant conditional on the level terms ($p = 0.44$); the naive test that omits the level terms rejects spuriously, because $\chi \cos 2\xi$ is nearly collinear with $\cos 2\xi$ (correlation 0.93). The second-order structure that exists is a level effect of about ± 9 percent along the east–west axis ($\cos 2\xi$: $+0.089$, $p < 0.001$), entered in the texture ledger.[1] An employment-weighted estimation leaves the depth harmonic intact ($m_5 = +0.848$, $R^2 = 0.704$).

[1]Since m_4 is indistinguishable from zero, the depth-priced core of the field is, in Cartesian coordinates, $\ln \Pi \approx m_0 + 0.89\, y$ plus the level terms: a single linear coordinate of the plane. We note this in passing; the model carries the full reduced form.

4.4 From centroids to bundles

The reduced-form step of Section 3.2 is from coefficients identified at occupational centroids to a field read at task locations, and eq. (3) makes it testable: the model prices an occupation as its bundle, not as the field at its centroid. The bundle-integrated wage predicts observed log wages with $R^2 = 0.510$ against 0.523 for the in-sample centroid evaluation (prediction R^2 without refitting; correlation 0.717), and the Jensen gap $\ln w_{\text{bundle}} - \ln \Pi(\mu_o)$ is small (mean +0.018, sd 0.049, range −0.168 to +0.195). Bundle dispersion matters little for pricing in the cross-section, which is what licenses estimating the field at centroids and operating it at tasks.

The same step crosses the support boundary stated in Section 3.2: the field is identified on centroids with $\chi \leq 0.752$, while tasks extend to the rim. The extrapolated region carries on average 7.3 percent of an occupation's bundle mass (median zero, maximum 66.7 percent; 7.5 percent of tasks unweighted). Its quantitative weight in pricing is negligible: clipping the field at the support boundary changes bundle log wages by +0.0007 on average (sd 0.0044, maximum 0.038) and moves the prediction R^2 from 0.5097 to 0.5101, while the extrapolated price at the northern rim exceeds the boundary value by 23 percent. The model uses the extrapolated field, consistent with the task layer being defined to the rim; the clipped variant is a sensitivity.

4.5 The price functional and the gate weights

The components p_k of the price functional, which the deficit gate of eq. (12) uses as weights v_k, are measured by replicating the global mediation regression of Storck and Andersson (2026) on the frozen inputs: log wage on the direction terms, χ, mean required education, and the four cluster intensities ($N = 785$, HC3, $R^2 = 0.683$). The replication is exact against the published values: χ −0.079 ($p = 0.24$; published −0.08, $p = 0.24$), S1 +0.328 ($p < 0.001$; published +0.33), S2 +0.049 ($p = 0.020$; published +0.05, $p = 0.02$), A1 −0.030 and A2 +0.010 (both insignificant). The direction terms are part of the replicated specification; the variant without them does not replicate. The gate accordingly runs on the priced clusters, $\mathcal{K} = \{\text{S1}, \text{S2}\}$ with $v = (0.328, 0.049)$, as derived in Section 3.5; the all-cluster variant is a sensitivity in the regime runs.

4.6 The wage wedge

The price field accounts for roughly half of the wage cross-section; the residual variance share is 0.477, with weak but significant spatial autocorrelation (Moran's I +0.080, permutation $p = 0.001$). The residual is not capability structure in disguise: the four cluster intensities explain 9.7 percent of it, while Job Family fixed effects explain 27.9 percent (33.4 jointly with the intensities, for a total R^2 of 0.682 against log wages). What remains above the field is mostly classification-level wage setting, which is precisely what the theory layer excludes and the measurement layer records. The occupation wedge is $\eta_o = \ln w_{\text{obs},o} - \ln w_{\text{bundle},o}$, and the family wedge η_g is its unweighted family mean, ranging from -0.261 (Personal Care and Service) to $+0.372$ (Management) and absorbing 30.2 percent of the occupation-level wedge variance (employment-weighted means correlate at 0.948). The wedge is measured once on this cross-section and held fixed under technology shocks. It enters the operated regime, when switched on, through the effective price $\Pi_{\text{eff}} = e^{\eta_g}\,\Pi$, so that dearer families cross the takeover margin sooner; the regime runs of Section 5 are reported with and without it.

4.7 Technology calibration

[TODO: Kalibrering. Det kognitiva fältet (p_K, z_K, A_K) kalibreras mot task-nivå-AI-exponering projicerad på geometrin (script 08, Eloundou et al. 2024 som primär källa; Gmyrek et al. 2025 endast om den direkta källan faller). Det manuella fältet placeras för hand (väst, smalt z_K, högt A_K för att korsa prisgrinden mot billig arbetskraft); öppen väg att binda det till RTI (autor2013) eller robotexponering (acemoglu2020robots), se noten i diskussionen. Karaktären sätts $s_K = 1$ i huvudmodellen. Attachment-skalan ℓ sätts tolkningsbart till inomriktnings-SD:n för det prissatta förmågeindexet. Efterfrågeelasticiteten η kalibreras inte utan sveps; i den historiska läsningen kan den sättas till Bessens skattade sektorselasticiteter. Enheter: R och Π i USD per timme, A_K dimensionslös effektivitet, τ marginbredd i samma enhet som $s_K\phi_K - R/\Pi$.]

5 Illustrative Cases

[TODO: Två teknikfält genom samma ekonomi (script 13). (1) Kognitivt: norr, brett, kalibrerat. (2) Manuellt/industriellt: väst, smalt, placerat. För vart och ett: svep η och visa sysselsättnings- och löneändring i fältets egna automatiserade region. Huvudfigur `demand_channel_eta.png`: %-ändring i regionens sysselsättning mot η (logskala), båda fälten, $\eta = 1$ markerat (reproducerar den kostnadsinvarianta modellen), rimligt elasticitetsband skuggat; båda friinställer vid trolig η och vänder först vid $\eta \approx 7$–8. Poängen är andra-ordnings-egenskapen: $\Pi/c \to 1$ vid adoptionsmarginalen (numeriskt 1.00), så besparingen och därmed efterfrågekanalen är svag där. Lägg till kartan över obundna uppgifter $u(\mathrm{r})$ färglagd med Π. Knyt till Bessens tre regimer (jordbruk oelastiskt $\to$ friinställning; 1800-talstextil elastiskt $\to$ tillväxt).]

6 Empirical Analysis

[TODO: Ingen DiD/event study (slopad: tröskeln z_K existerar inte under reading A, och ChatGPT-eventet bär inte den kausala vikten). Kvar står konsistenstestet av förträngningskanalen: projicera centroidförskjutningen $\Delta\mu_o$ på prisgradienten $\nabla\Pi$ och pröva mot observerade löneändringar 2019–2024 (OEWS, samma ~741 yrken). Uttryckligen icke-kausalt: en riktnings-konsistens kontroll av om takeover-inramningen pekar rätt, inte en identifierad effekt. Överväg ett enkelt tvärsnittstest av Bessen-mekanismen: korrelera observerad sektorssysselsättningsförändring under mekanisering med skattad efterfrågeelasticitet, som extern plausibilitet för η-kanalen.]

7 Discussion

The central mechanism is the directional pricing of work. Reinstatement is good for employment wherever it lands, because it returns task mass to labor. It is good for wages only where the price field rewards the work it returns. New tasks seeded into the analytical directions, where depth is priced, can raise both employment and the wage. The same new tasks seeded into the service and manual directions raise employment but not the wage, because depth is unpriced there. A reinstatement elasticity that looks protective in aggregate can therefore leave manual workers with more work and no more pay. This is a distributional statement that a one-dimensional model cannot make, because it has no direction in which to place the new work.

Default automation and default reinstatement pull the wage in opposite directions, and the price field decides the contest. The first term of (16) strips the dear tasks, since the operated threshold (6) is crossed first where Π is high. The second term refills with tasks that the deficit gate admits, which are mostly cheap. Left to run, the process bleeds priced work and returns unpriced work, the bundle form of the Turing trap. The same price channel supplies its own brake. As a bundle loses priced mass to capital, its average price falls, which raises the threshold R/Π for the work that remains and slows further takeover in that direction. This is the self-correction of Acemoglu and Restrepo (2018), where a wave of automation lowers the effective cost of the labor that is left and discourages further automation, here inherited through the price field rather than assumed.

Whether automation raises or lowers employment in a direction is a separate question from the wage, and the demand for output decides it. The takeover lowers the unit cost of the work it touches, and under competition that saving reaches buyers as a lower price. Where demand for the cheapened output is elastic, the quantity rises enough to draw labour into the automated direction; where it is inelastic, the price falls without a matching rise in quantity and labour is released. This is the demand mechanism of

Bessen (2019), here resolved by direction: the same field grows employment in an elastic direction and sheds it in an inelastic one, and the geometry adds the location at which each happens. But the price gate bounds how strong the channel can be. Because capital is adopted only where it is the cheaper producer, the saving it realises is second-order at the adoption margin, vanishing as the operated share approaches one half and growing only deep in the core, the spatial form of the second-order automation margin of Acemoglu and Restrepo (2018). The demand channel is therefore weak near the margin, and at plausible elasticities the first-order displacement of the takeover dominates: net employment growth in an automated direction requires either demand elastic enough to overcome the labour saving or automation deep enough to realise a large per-unit saving. The historical record sits on both sides of this condition. Mechanised agriculture, with inelastic demand for food and a modest per-unit saving, shed labour; nineteenth-century textiles, with a large cumulative saving and demand elastic enough that consumption outran the labour saved, grew employment for decades before demand saturated (Bessen 2019). Where the demand for cognitive output falls on this scale is what the consequences of cognitive automation turn on.

The displacement falls within occupations, not between regions, because it lives on the interior margin $a(\mathbf{r})$ rather than on a boundary in space. A bundle spans a range of locations, and the operated share varies across them, so some of an occupation's own tasks cross the margin while others do not. This is why the occupation is the wrong unit and the task is the right one. The point has an empirical counterpart: the automatability of work varies more within occupations than between them (Brynjolfsson, Mitchell, and Rock 2018), which is the same statement as a bundle spanning a range of ϕ_K. The within occupation margin is therefore not a modeling convenience but the level at which the displacement actually occurs.

Exposure and incidence are distinct, and the model keeps them apart. The field ϕ_K is technical exposure, what capital can do at a location. The operated share $a(\mathbf{r})$ is economic incidence, what capital is paid to do there, set by exposure against price through (6). The two need not coincide. A weak correlation between exposure and wages does not weaken the trap, because the trap is a claim about where operation is profitable given exposure, not about where exposure happens to fall. Directed implementation concentrates on the work that is both exposed and dear (Brynjolfsson, Mitchell, and Rock 2018), which is exactly the high-Π high-ϕ_K region where (6) is crossed first. Reading exposure alone would miss this; the price field is what turns exposure into incidence.

The deficit gate is also the boundary between two outcomes that the debate often conflates. Where attachment capacity $C(\mathbf{r})$ is positive, the bundle is redrawn and the occupation is redefined. Where $C(\mathbf{r})$ collapses, the seeded mass stays unbound, and where it clusters it marks a candidate occupation that no existing bundle can hold. Redefinition and the birth of a new occupation are then the two sides of the same gate, separated by whether any occupation already holds the priced capability the new work requires.

Several limitations bound these claims. The worker layer is closed as a static equilibrium, so the results are comparative statics between two rest points rather than a path in time: the adjustment dynamics — reading the same model as laws of motion in calendar time, with the transitional mismatch between displacement and reinstatement as the central object — are the subject of ongoing work, as is the response of total labor supply to aggregate value. One consequence is that the rise-and-fall of employment that Bessen (2019) documents over time, as a sector's demand elasticity falls with saturation, appears here only as a cross-section over η, not as a trajectory; recovering it as a trajectory is left to that dynamic extension. The output market enters only through the demand multiplier D: each place faces an independent isoelastic demand, with no cross-place substitution and no income effects, and the elasticity η is swept rather than estimated, so the framework remains partial and its employment statements are conditional on η. New tasks enter as added weight on existing locations rather than as genuinely new points on the disk, which understates the possibility that automation opens directions the geometry does not yet contain. The cluster weights v_k and the acquisition scale ℓ are identified from the cross-section and held fixed, and the wage test of the displacement channel, projecting the centroid shift $\Delta\mu_o$ on $\nabla\Pi$ against observed wage changes, is the sharpest available check on whether the takeover framing is right.

[TODO: Relatera till Autor & Thompson (2025) när referensen verifierats; placera i stycket om riktningsberoende prissättning eller gränszonen.]

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